

# IC platform test configuration and reporting guide

This document explains the new `lex_test_config.yaml` workflow, how the Instance Controller Platform UI uses it, how local report generation and email sending work, and what teams need to configure in practice.

## 1. Overview

The testing flow is now built around a repository-level configuration file called `lex_test_config.yaml`. This file defines the default setup for test execution, PDF reporting, and report email delivery.

Tests can now generate branded PDF reports and send emails with the report attached. This makes it easier to share results with the relevant recipients while keeping the configuration in the repository instead of repeating it manually for every run.

## 2. How the test handling works

The feature is implemented as a wrapper around **pytest**. In practice, `python -m lex pytest` loads the effective `lex_test_config.yaml`, adds the configured test entrypoint when needed, and then forwards the actual test run to pytest.

### Lex-specific flags

- `--report` generates the PDF report only.
- `--report-and-email` generates the PDF report and prepares or sends report emails.

### Standard pytest behavior still applies

Everything else goes through to pytest unchanged, including marker expressions, keyword filtering, file paths, and direct node IDs.

**NB:** Lex is wrapping pytest, so if you need more detailed information about filtering, node selection, markers, or other advanced behavior, refer to the pytest documentation.

### 3. What `lex_test_config.yaml` does

The YAML file defines the default test configuration for a repository.

| Field                          | Purpose                                                  | Example                                                        |
|--------------------------------|----------------------------------------------------------|----------------------------------------------------------------|
| <code>tests_entrypoint</code>  | Tells Lex where pytest should start collecting tests.    | <code>Tests</code>                                             |
| <code>receivers</code>         | Default recipients for reports.                          | <code>QA</code>                                                |
| <code>report.output_dir</code> | Folder where PDF reports are written.                    | <code>reports</code>                                           |
| <code>email.*</code>           | Sender name, sender email, reply-to, and subject prefix. | <code>Lex Reports</code> ,<br><code>noreply@example.com</code> |
| <code>groups</code>            | Logical test groups users can run and report on.         | <code>smoke</code> , <code>regression</code>                   |

#### Use the example file as a starting point

The attached `lex_test_config.yaml.example` can be copied into your repository and renamed to `lex_test_config.yaml` . After that, adjust the defaults for your project.

Global `receivers` always receive all configured group results. If a group defines its own `receivers` , those addresses are added for that group and do not replace or suppress the global receivers.

```

tests_entrypoint: Tests

receivers:
  - name: "QA"
    email: "qa@example.com"

report:
  output_dir: reports

email:
  from_email: "noreply@example.com"
  from_name: "Lex Reports"
  reply_to: "noreply@example.com"
  subject_prefix: "Lex test report"

receivers:
  - name: "QA"
    email: "qa@example.com"

groups:
  - name: smoke
    description: "Fast checks safe to run frequently."
    receivers:
      - name: "Smoke owner"
        email: "smoke-owner@example.com"

  - name: regression
    description: "Broader verification before release."

```

## 4. Defaults versus UI overrides

- `lex_test_config.yaml` defines the repository defaults.
- When you start a test from the **Instance Controller Platform** workflows list, those values are loaded into the UI automatically.
- You can override the loaded values in the UI for that run only.
- When UI values are present, they take priority over the YAML defaults.

## 5. What users need to do

1. Take the attached `lex_test_config.yaml.example`, copy it into your repository, and rename it to `lex_test_config.yaml`.
2. Define the default test entrypoint, report recipients, report output folder, email settings, and the test groups you want to expose.
3. Commit the file to the repository.

4. In the Instance Controller Platform, start a test from the workflows list and review the preloaded values in the UI.
5. Override values only when needed for that specific run.

## Pytest markers for grouped tests

For group-based selection to work properly, the relevant tests should use matching pytest markers. Typical examples are `@pytest.mark.smoke` or `@pytest.mark.regression`. If an entire file belongs to the same group, you can also use module-level marking such as `pytestmark = pytest.mark.smoke`.

## 6. Local usage

`python -m lex pytest` can be used locally. When the repository contains `lex_test_config.yaml`, Lex loads it automatically and applies its defaults.

```
# Run all tests from the YAML tests_entrypoint
python -m lex pytest

# Run only one group
python -m lex pytest -m smoke

# Run multiple groups
python -m lex pytest -m "smoke or regression"

# Run a specific file
python -m lex pytest Tests/example/test_example.py

# Run a single test
python -m lex pytest Tests/example/test_example.py::TestExample::test_case

# Filter by keyword
python -m lex pytest -k ranking

# Inspect discovered groups and tests without running them
python -m lex pytest-groups
```

## 7. Local PDF reporting and email sending

The following commands are the main entry points for local report generation and email sending:

```
# Generate the PDF report locally  
python -m lex pytest --report
```

```
# Generate the PDF report and confirm email sending interactively  
python -m lex pytest --report-and-email
```

## SendGrid setup for local email sending

To send emails locally, you need a SendGrid API key. The key will be distributed separately by email. Add it to your project `.env` file like this:

```
SENDGRID_API_KEY=your_sendgrid_api_key
```

## Available sender addresses

- `tests@lund-consulting.de`
- `noreply@scaled.pe`

If you would prefer another sender to be configured, please reach out by email.

## 8. Workflow reliability improvements

In addition to the new configuration and reporting flow, the release also includes workflow-related fixes, including GitHub-triggered workflow fixes, a fix for purge after spin-up, and more stable workflow log handling.

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This PDF is intended to be attached to the release email as the detailed reference for teams.